MILITARY SPECIFICATION

HOSE ASSEMBLIES, CABLE ASSEMBLIES, COMMUNICATION, ASSOCIATED

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the cable assembly described herein shall consist of this document and the issue in effect of Specification MIL-H-81581.

1. SCOPE

1.1 This specification covers the requirements for communication cable assemblies used with the applicable hose assemblies utilized in the Naval aviator's life support systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods

Military

MIL-P-116	Preservation, Methods of
MIL-W-16878/4	Wire, Electrical Type E, 200° C and 260° C, 600 Volts, (Insulated, High Temperature) (Navy)
MIL-C-45662	Calibration of Standards

STANDARDS

Federal

FED-STD-595	Colors

FSC 1660

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STANDARDS

Federal (Continued)

FED-STD-601	Rubber, Sampling, and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-794	Parts and Equipment Procedure for Packaging and Packing

DRAWINGS

Naval Air Systems Command

283AS100	Cable Assemblies Index, Communications

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the Government quality assurance representative. Copies of this specification and other unclassified specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Standard

D1149	Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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3. REQUIREMENTS

3.1 Requirements - Requirements shall be in accordance with the general specification MIL-H-81581 and as specified herein.

3.2 Qualification - The cable furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening for bids. Unless otherwise specified, each cable assembly listed on the NASC Dwg. 283AS100 shall be qualified separately.

3.3 Materials and components -

3.3.1 Elastomeric material - The elastomeric material used in the fabrication of the moded or extruded portions of the cable assembly shall be of a composition which shall not bloom, shall have a minimum Shore Durometer Hardness of 55 ±5 when tested as specified in Method 3021 of FED-STD-601, and shall meet the ozone resistance requirements of the general specification.

3.4 Design and construction - The communication cable assembly shall be in accordance with the applicable drawing listed in NASC Dwg. 283AS100 and as specified herein.

3.4.1 Communication cable - The communication cable shall consist of insulated conductors, a restraint cord (if applicable), a protective covering and termination in accordance with the applicable drawing listed in NASC Dwg. 283AS100.

3.4.1.1 Communication conductors - Each insulated communication conductor shall consist of 19 silver coated copper strands (No. 36 AWG) bunched with a lay of not more than 3/4 inch. The conductor strands shall be uniformly coated in accordance with MIL-W-16878/4 and the conductors shall have such termination as specified on the applicable drawing. The communication conductors shall be twisted with a lay or lays, which shall not exceed 1-3/4 inches.

3.4.1.2 Conductor insulation - A uniform covering of polytetrafluoroethylene insulating material shall be placed over each conductor in accordance with MIL-W-16878/4. For circuit and polarity identification, the insulation of each conductor shall have a different distinctive color throughout its length as specified on the applicable drawing.

3.4.1.3 Conductor shielding - Conductor shielding shall be as specified on the applicable drawing and shall provide a coverage from 85 to 95 percent. Shielding shall consist of tin-plated soft drawn copper (No. 40 AWG to AWG).

3.4.1.4 Electrical fitting - When the applicable control drawing specifies the attachment of electrical fittings, such as jacks, plugs, terminal lugs, etc., the

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electrical fittings shall be attached to the cable in a manner which will in no way impair the designed performance of the electrical fittings or the cable assembly. Cable assemblies having electrical fittings may be tested for insulation resistance with or without the electrical fitting attached, at the discretion of the test activity. However, when tested with the electrical fittings attached, the specification requirements of the electrical fitting shall govern in the event of conflict between the insulation resistance requirements specified herein and the requirements of the specification for the electrical fitting.

3.4.1.5 Restraint cord - The restraint cord, when specified on the applicable control drawing, shall be of a polyamide or polyester construction.

3.4.1.6 Protective covering - The protective covering shall be constructed of a concentric inner and outer layer of materials. The inner layer shall consist of a fibrous sheathing, constructed of a synthetic compound, which shall be closely wrapped, braided, or applied as a continuous tube around the shielded, twisted, insulated conductors and restraint cord (when specified). The fibrous sheathing shall in turn be tightly contained by a molded or extruded outer layer of elastomeric material which shall meet the requirements of 3.3.1. The protective covering shall be of uniform thickness and shall continuously cover the length of cable as specified on the applicable drawing.

3.5 Performance characteristics -

3.5.1 Restraint cord elongation and tensile strength - When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds when tested in accordance with 4.4.2.

3.5.2 Cable tensile strength - When tested as specified in 4.4.3 the cable assembly shall be capable of withstanding a tensile load of 35 pounds with no evidence of damage to the cable.

3.5.3 Insulation resistance - Unless otherwise specified in the applicable control drawing, the insulation resistance of each electrical circuit shall be 100 megohms minimum when tested as specified in 4.4.4.

3.5.4 Polarity (color code) and electrical continuity - The cable assembly, when tested as specified in 4.4.5, shall pass the polarity (color code), and in each case the electrical resistance of each circuit contained in the cable assembly shall be not greater than 0.2 ohm, plus 0.03 ohm per foot length of cable.

3.5.5 Ozone resistance - When tested as specified in 4.4.6 the test specimens shall show no evidence of checking, cracking, or other damage when examined under a 10X magnification.

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3.5.6 Low temperature - The cable assembly shall meet the requirements of 3.5.3 and 3.5.8 when tested as specified in 4.4.7.

3.5.7 High temperature - The cable assembly shall meet the requirements of 3.5.3 and 3.5.8 when tested as specified in 4.4.8.

3.5.8 Flexibility - The cable assembly shall have no noticeable permanent set or other damage when tested as specified in 4.4.9.

3.5.9 Flexibility endurance - The cable assembly shall be capable of being flexed without loss of conductor continuity or other damage when tested as specified in 4.4.10.

3.6 Color - Unless otherwise specified, the color of the elastomeric portions of the cable assembly shall be uniformly green, approximately matching color number 34079 of FED-STD-595.

3.7 Soldering - Soldering shall conform to MIL-STD-454 requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Test equipment and inspection - Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Inspection conditions - Unless otherwise specified herein, all inspections shall be made at room ambient temperature, relative humidity and pressure.

4.3 Sampling and inspection - Sampling and inspection shall be in accordance with the general specification and as specified herein.

4.3.1 Qualification inspection - The qualification inspection of the cable assembly shall consist of examinations and tests for all the requirements of this specification.

4.3.1.1 Qualification samples - The qualification inspection samples shall consist of the following:

(a) Four cable assemblies for which qualification is required.

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(b) Three test slabs 6 by 6 by 0.075 inches composed of the identical ozone-resistance composition that was used in manufacturing the outer layer of the protective covering of the cable assembly.

(c) Ten 30 inch long restraint cord specimens.

4.3.2 First article inspection - The first article inspection of the cable assembly shall consist of the examinations and tests specified in Table I.

4.3.2.1 First article samples - The first article sample shall consist of four cable assemblies of the part number contracted for.

TABLE I

FIRST ARTICLE INSPECTION

INSPECTION	REQUIREMENT	METHOD
Visual examination	-	4.4.1.1
Flexibility	3.5.8	4.4.9
Cable tensile strength	3.5.2	4.4.3
Insulation resistance	3.5.3	4.4.4
Polarity and electrical continuity	3.5.4	4.4.5
Low temperature	3.5.6	4.4.7
High temperature	3.5.7	4.4.8
Flexibility endurance	3.5.9	4.4.10
Preparation for delivery	-	4.4.1.2

4.3.3 Quality conformance inspection - The quality conformance inspection shall consist of the inspections specified in Table II.

4.4 Inspection methods -

4.4.1 Visual examination -

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TABLE II

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE CABLE ASSEMBLIES

INSPECTION	REQUIREMENT	METHOD	SAMPLE SIZE 1/	ACCEPTANCE CRITERIA 2/
Visual examination (See classification of defects)	-	4.4.1.1	Every cable assembly for critical defects. Inspection Level II for minor defects	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per 100 units for minor defects.
Flexibility	3.5.8	4.4.9	Inspection Level S-2	An acceptable quality level of 1.5 defects per 100 units.
Cable tensile strength	3.5.2	4.4.3	Inspection Level S-2	An acceptable quality level of 1.5 defects per 100 units.
Insulation resistance	3.5.3	4.4.4	Inspection Level S-2	An acceptable quality level of 1.5 defects per 100 units.
Polarity and electrical continuity	3.5.4	4.4.5	Every cable assembly	Reject all defective units.
Preparation for delivery	-	4.4.1.2	Inspection Level S-2	An acceptable quality level of 4.0 defects per 100 units.

1/ Sampling for acceptance inspection shall be in accordance with MIL-STD-105.

2/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

4.4.1.1 Cable assembly - Every cable assembly shall be examined visually (for critical defects) to determine conformance to this specification and applicable drawings. The classification of defects, Table III, shall be used to classify the defects found, as applicable.

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TABLE III

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE CABLE ASSEMBLY

CRITICAL	MINOR
1. Dimensions not within specified tolerances. 2. Material imperfections - foreign matter embedded. 3. Any component missing, malformed, fractured, or otherwise damaged. 4. Any component loose, or otherwise not securely retained. 5. Incorrect assembly or improper positioning of components. 6. Any functioning part that works with difficulty. 7. Faulty workmanship or other irregularities.	201. Marking - missing insufficient, incorrect, illegible, or not permanent. 202. Color not as specified.

4.4.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing cable assemblies of one type, selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects Table IV, shall be used to classify the defects found.

TABLE IV

CLASSIFICATION OF DEFECTS FOR PREPARATION FOR DELIVERY

ITEM	DEFECTS
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence of method of application; markings not the same on the interior and exterior containers.

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TABLE IV (Continued)

ITEM	DEFECTS
Packaging and packing materials	Any nonconforming components; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate packing, container flaps, or loose strapping, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirement.

4.4.2 Restraint cord elongation and tensile strength - The restraint cord shall be tested in accordance with CCC-T-191 Method 4102. The restraint cord elongation and tensile strength shall conform to the requirements of 3.5.1.

4.4.3 Cable tensile strength - The cable assembly shall be subjected to a tensile load of 35 pounds applied at the ends of the cable acting along the longitudinal axis. The tensile load shall be maintained for one minute. The cable assembly shall conform to the requirements of 3.5.2. After completion of this test the cable assembly shall be subjected to and meet the applicable requirements of the tests specified in 4.4.4.

4.4.4 Insulation resistance - This test shall be as specified in the general specification.

4.4.5 Polarity (color code) and electrical continuity - This test shall be conducted as specified in the general specification.

4.4.6 Ozone resistance - This test shall be conducted as specified in the general specification.

4.4.7 Low temperature - The cable assembly shall be conditioned at -65 ±5° F for 4 hours. After the conditioning period and while still at this temperature, the cable assembly shall be subjected to the tests specified in 4.4.9. The cable assembly shall then be conditioned at room temperature and subjected to the tests specified in 4.4.4 and 4.4.5.

4.4.8 High temperature - The cable assembly shall be conditioned at 160 ±5° F for 4 hours. After the conditioning period and while still at this temperature, the cable assembly shall be subjected to the tests specified in 4.4.9. The

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cable assembly shall be returned to room temperature and subjected to the tests specified in 4.4.4 and 4.4.5.

4.4.9 Flexibility - The cable assembly shall be closely coiled about a rod 1 inch in diameter and then released. The test shall be repeated four (4) times. After each release, the cable assembly shall meet the requirements of 3.5.8.

4.4.10 Flexibility endurance - The cable assembly shall be coiled about a rod one (1) inch in diameter and released. This operation shall be repeated for 500 cycles. After this test, the cable assembly shall be subjected to the tests specified in 4.4.4 and 4.4.5. The cable assembly shall conform to the requirements of 3.5.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A in accordance with MIL-STD-794 utilizing Method 1C-3 of MIL-P-116 without a preservative.

5.2 Packing - Packing shall be level A, B, or C as specified (see 6.2 of general specification) and in conformance with MIL-STD-794.

5.3 Marking - Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The cable assembly covered by this specification is intended for use for radio services to aircrewmen in the operation of aircraft.

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